

EMERGING TECHNOLOGIES: AN OVERVIEW AND KEY AREAS

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1. Introduction

1.1 Background of the Module

Module 1 introduces emerging technologies by discussing how technologies develop and become important in real-world settings. One important point in the lesson is that an emerging technology is not always something that has just been invented. A technology can still be considered emerging when it continues to improve, is applied in new ways, or becomes involved in different areas of research. The module also discusses some of the possible applications of these technologies in different settings.

1.2 Learning Objectives

The module has three main objectives. Students are expected to explain what emerging technology means, describe its different areas, and analyze several emerging technologies. These objectives show that the lesson is not only focused on learning definitions. It also requires students to understand how technologies develop and how they can be applied in different situations.

2. Overview of Emerging Technologies

2.1 Definition of Emerging Technology

Emerging technology is not merely about technology that has just been newly invented. Instead, it refers to technology that is still being developed and may bring changes to business and society. These technologies may have already been used for years but continue to evolve as people improve them and discover new ways to apply them. Because of this, the age of a technology is not enough to determine whether it is emerging. As mentioned in the module, artificial intelligence is one example of an emerging technology. Despite being developed in the 1950s, it is still considered emerging because it continues to develop and is being used in various fields. This shows that technological emergence is an ongoing process and is not simply a description of technology that is newly invented.

3. Characteristics of Emerging Technologies

3.1 Radical Novelty

Radical novelty refers to the newness of a technology or the use of an existing technology in a new way. This means that a technology does not always have to be newly created to show radical novelty. It can also perform a similar function using a different basic principle compared to the technology used before it. Artificial intelligence is one example mentioned in the module because AI can be applied in different ways to achieve different results. This shows that newness can also come from finding new uses for technologies that already exist.

3.2 Relatively Fast Growth

Relatively fast growth was the growth seen in the emerging technology in several areas, such as the number of people or organizations involved, funding, knowledge produced, prototypes, products, and services. AI is used as an example because its users continue to grow, including companies, universities, and individuals. This means that technological growth is not only about how many people use a technology. It can also be seen through research, funding, development, and the creation of new products and services.

3.3 Coherence

Coherence is about technologies or research areas becoming connected instead of remaining separate. The lecture explains that emerging technology can involve the convergence of technologies that have already moved beyond being purely conceptual. These connections also need to persist over time for the technology to develop a clearer identity and momentum. One example is the grouping of cloud computing, AI, data analytics, and robotics for different uses. This shows how separate technologies can become connected as they continue to develop together.

3.4 Prominent Impact

An emerging technology can affect more than just one specific area. The lecture explains that its effects can reach different sectors and may influence industries, economic activity, institutions, interactions, and knowledge production. AI is one example because it is being used in businesses, academe, healthcare, and by individuals around the world. Because of this, the impact of an emerging technology can be seen in the different areas where it is being used and the changes it can bring.

3.5 Uncertainty and Ambiguity

There is still a lot that is not known about emerging technologies, especially when it comes to their future uses and effects. The lecture explains that some of their most significant impacts may happen in the future, while their possible applications can still change as the technology develops. AI is an example of this because there are many possible ways it could be used, but not all of these applications are guaranteed to be successfully developed. People may also have different ways of understanding or interpreting the same technology. Because of this, it is difficult to know exactly how an emerging technology will develop and what effects it will have in the future.

3.6 Synthesis of the Five Characteristics

These five characteristics help explain why a technology can be considered emerging. It may show newness, grow quickly, become connected with other technologies, create effects across different sectors, and still have uncertain future uses. The lecture especially emphasizes that its

most prominent effects may happen in the future. A technology can therefore already have useful capabilities while its wider applications and long-term effects are still developing.

Characteristic	Main idea	Lecture Example	Analysis
Radical novelty	Newness through progress or new uses	Different applications of AI	Emergence can result from significant new uses.
Relatively fast growth	Rapid expansion across several dimensions	Growing AI users	Growth can involve users, research, funding, prototypes, products, and services.
Coherence	Convergence of connected technologies	Cloud, AI, data analytics, robotics	Separate technologies can form a connected group with shared momentum.
Prominent impact	Broad influence across sectors	AI in business, academe, healthcare, and among individuals	Emerging technologies can change industries and interactions.
Uncertainty and ambiguity	Future outcomes and uses remain unsettled	Possible AI applications may not be successfully developed	Emergence involves incomplete knowledge and flexible applications.

4. Areas of Emerging Technologies

4.1 Augmented Reality (AR)

Augmented Reality, or AR, is a technology that adds computer-generated displays, text, sound and effects to what a person sees in the real world. The real environment is still visible, but digital elements are added to it. The module gives several examples of this, including an in-car display that can guide a driver and help with reverse parking, Pokémon GO and applications

that allow shoppers to see how a piece of furniture would look in their living room before buying it. These examples show that AR is not limited to one type of activity since it can be used in driving, entertainment, and shopping.

4.2 Artificial Intelligence (AI)

Artificial Intelligence, or AI, is described in the module as the simulation of human intelligence processes through machines. These include learning, reasoning, and self-correction. AI is also one of the technologies used throughout the lesson to explain what makes a technology emerging. Even though AI was developed in the 1950s, it is still developing and being used in different areas. Because of its continued development and increasing applications, the module uses AI as an example when discussing several characteristics of emerging technology.

4.3 Machine Learning

Machine learning is an application of AI that gives machines the ability to learn from experience in a way similar to humans. The module shows that it can be applied in healthcare, financial services, marketing services and transportation. One example that stood out in the lesson is its use in healthcare, where machine-learning algorithms can be used to identify and develop new medicines more quickly. This shows how machine learning can take the idea of a machine learning from experience and apply it to an actual problem in a specific field.

4.4 Quantum Computers

Quantum computers are another emerging technology discussed in the module. Unlike conventional computers, they use quantum mechanics when performing calculations and may be able to solve some problems much more efficiently. The basic unit of a quantum computer is a qubit. While a standard bit can be 0 or 1, a qubit can exist in a superposition of the two computational states. The module mentions IBM, Google, Rigetti Computing and IonQ in connection with quantum computers demonstrated in laboratories. It also discusses possible future applications, such as simulating atoms and molecules and helping with the design of materials.

5. The Fourth Industrial Revolution

5.1 Industrial Revolution and Production Technology

The Industrial Revolution refers to the different stages in which advances in technology changed the way production was carried out. The module describes it as a systematic transformation of manufacturing through the integration of physical and digital systems. These changes aim to improve quality, reduce costs, and increase efficiency. They also affected people's working conditions and lifestyles as production technologies continued to develop. The four stages show how production progressed from mechanical methods toward more automated, connected, and digitally integrated systems.

5.2 Industry 1.0

Industry 1.0 began in the eighteenth century and marked the move toward mechanical production. The module associates this stage with the use of hydroelectric and steam-powered equipment. Compared with production methods before this period, the introduction of these technologies allowed machines to take a greater role in production. Industry 1.0 can therefore be seen as the beginning of the changes in production technology discussed throughout the four industrial revolutions.

5.3 Industry 2.0

Industry 2.0 began in the nineteenth century with the introduction of technological systems that used electricity. Electricity allowed production to increase and supported the development of more sophisticated machines. This stage continued the changes brought by mechanical production in Industry 1.0, but expanded production capabilities through the use of electrical technology.

5.4 Industry 3.0

Industry 3.0 began with the first computer era and introduced a stronger role for electronics and information technology in production. These technologies were used to automate production further, while Internet access and connectivity contributed to advances in manufacturing and automation. The module also connects this stage with renewable energy. This period shows how production was gradually moving beyond mechanical and electrical systems toward the use of computers and information technology.

5.5 Industry 4.0

Industry 4.0 is the current and still developing stage of the Industrial Revolution. The module describes it through technologies such as the Internet of Things, robotics, Virtual Reality, Augmented Reality, and Artificial Intelligence. These technologies are changing how people live and work while also changing how production systems operate. Industry 4.0 can affect the relationships between suppliers, producers, and customers, as well as the interaction between humans and machines. Unlike the earlier stages, it places a stronger focus on connecting physical and digital systems.

Stage	Development	Major Change
Industry 1.0	Mechanical production using hydroelectric and steam-powered equipment	Mechanization

Industry 2.0	Electricity-based technological systems	Greater production and more sophisticated machines
Industry 3.0	Computers, electronics, IT, Internet access, connectivity, renewable energy	Further automation and connectivity
Industry 4.0	IoT, robotics, VR, AR, AI in a current and developing environment	Changing production relationships and human-machine interaction

6. Building Blocks of Industry 4.0

6.1 Big Data Analytics

Big Data Analytics deals with large amounts of data collected from different sources. Before the data can be useful, it needs to be collected and organized, then analyzed to find patterns and other information that can help a system. In Industry 4.0, this is particularly useful for making decisions in real time. The module gives production and customer-management systems as examples where data analysis can be applied. When large amounts of data can be processed and useful results can be obtained quickly, decisions can be made based on information about what is currently happening.

6.2 Internet of Things (IoT)

The Internet of Things, or IoT, refers to physical objects being connected to the Internet so they can communicate and interact with other devices. Sensors and wireless technologies help make this communication possible. Instead of devices operating separately, they can share information and become part of a connected system. In Industry 4.0, IoT can allow field devices to communicate with more centralized controllers. Some analytics and decision-making can also be handled closer to the devices instead of relying on one central location. This allows the system to respond to information in real time and produce results more quickly.

6.3 Cloud Computing

Cloud computing refers to the delivery of hosted services through the Internet. In Industry 4.0, it becomes useful as machine data and certain machine functions are increasingly placed in a connected cloud environment. The module also mentions that improvements in cloud performance can allow systems to react within several milliseconds. With machine information available through the cloud, it can be used together with other digital services instead of

remaining separate. This can support more data-driven services in production and connect machine data with the wider digital environment of Industry 4.0.

6.4 Augmented Reality in Industry 4.0

Augmented Reality, or AR, can also be used in industrial work and is not limited to entertainment or consumer applications. The module gives examples such as using AR to help workers select parts in a warehouse and providing repair instructions through mobile devices. Having this information available while the worker is doing a task can help with decision-making and following the proper work procedure. However, the module explains that these applications are still in their early stage. As AR continues to develop, companies may be able to use it more widely to provide workers with real-time information during different production activities.

7. Conclusion

Module 1 explains that emerging technologies are not always technologies that have just been invented. They can also be technologies that continue to develop, are used in new ways, and may bring changes to different areas. The five characteristics discussed in the module help describe how these technologies grow, connect with other technologies, create an impact, and remain uncertain in terms of their future uses. The module also introduces Augmented Reality, Artificial Intelligence, Machine Learning, and Quantum Computers and discusses some of their applications and possibilities. The four stages of the Industrial Revolution show how production has developed from mechanical systems to the more connected and digitally integrated environment of Industry 4.0. Finally, Big Data Analytics, IoT, Cloud Computing and AR show how different technologies can work together as part of this connected environment, while their continued development may lead to further applications in the future.

8. References

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